ENGR 111
Introduction to Engineering
Class #3

The engineering design process

How engineering actually gets done

What engineers get paid

Median annual wage, May 2024. Across all US workers the median was $49,500.

Computer hardware. $155,020. The top of the list, is this number trending up or down in 2026?

Aerospace $134,830. Chemical $121,860. Electrical $111,910.

Mechanical $102,320. Industrial $101,140. Civil $99,590.

The spread inside engineering is about one and a half times, top to bottom. The gap between this list and the national median is more than double.

Choose a discipline for the work. The differences within this list are smaller than the difference between it and everything else.

How good was the 2015 forecast?

In 2015 the Bureau of Labor Statistics published projections for 2024. Here is how they did.

Mechanical. Predicted 292,100 by 2024. Actual, 286,760. Almost exact.

Civil. Predicted 305,000. Actual, 355,410.

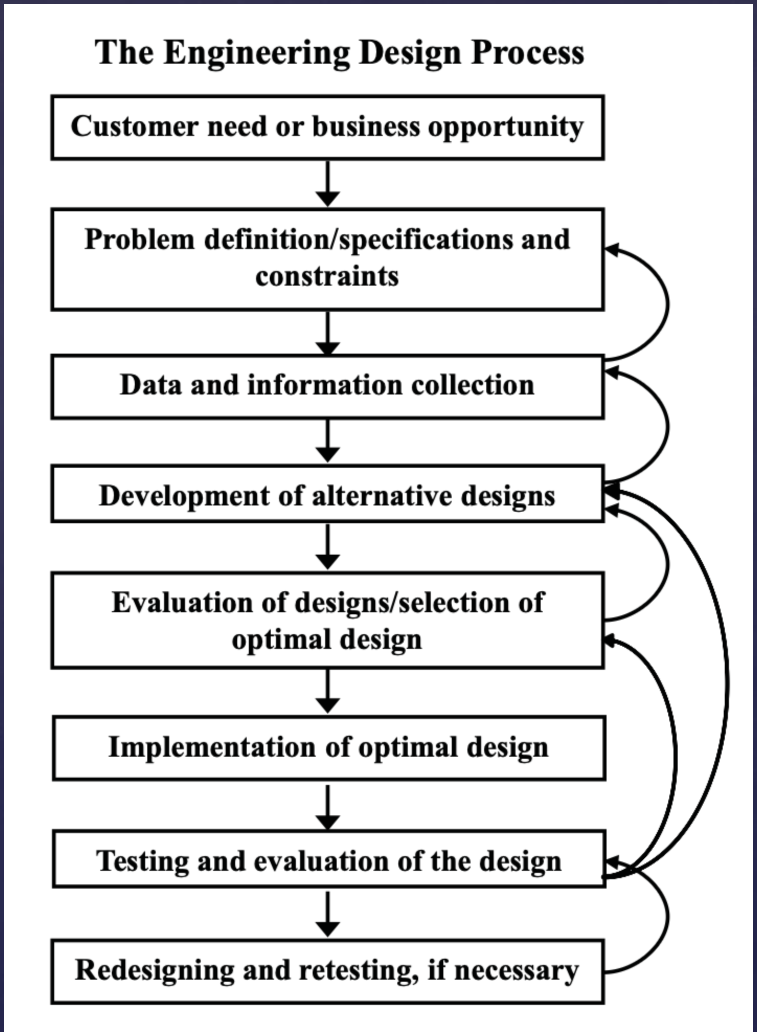
Industrial. Predicted 243,200. Actual, 350,230. Off by more than a hundred thousand.

Chemical. Predicted 34,900. Actual, 21,600. Wrong in the other direction.

Forecasting is difficult. Current figures are at bls.gov and they move every year.

The engineering design process

The model in your textbook. It is the common one in engineering, but it is not the only one.



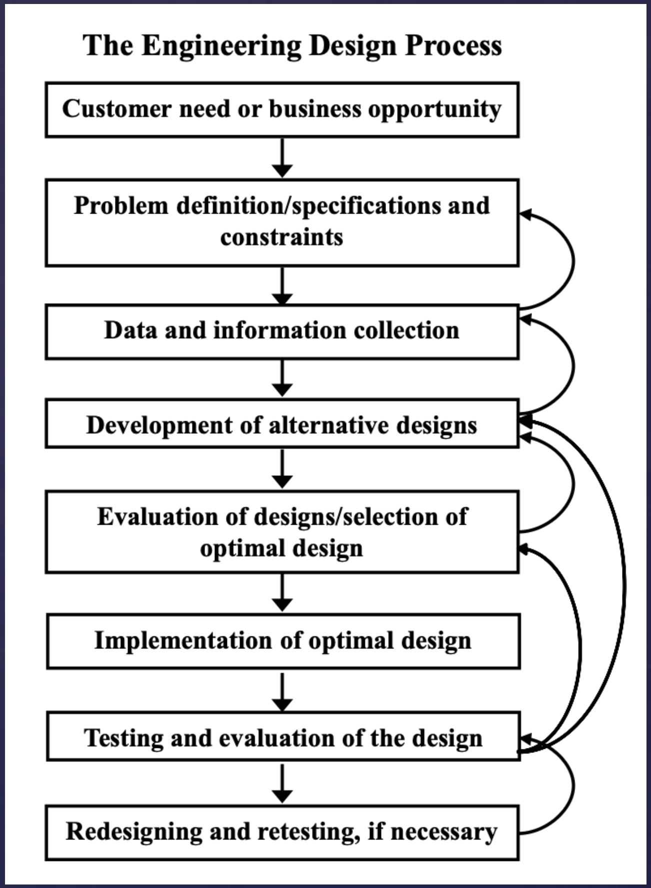
Steps 1 to 3: from a need to a plan

You are not choosing a design in any of these. That is not a delay, it is the job.

Customer need or business opportunity. Somebody has a problem, or you notice an opportunity nobody has asked about yet.

Problem definition, specifications and constraints. Constraints can be as ordinary as which parts, materials, people or facilities you can actually get hold of. The final design has to satisfy all of them.

Data and information collection. Go and get the numbers you do not have. Count how long the queue actually is. Test three candidate materials. Look up the standard. Where the number has never been published, build a small rig and measure it. Measuring, not choosing.



If you cannot say what problem you are solving, you are not ready to design anything.

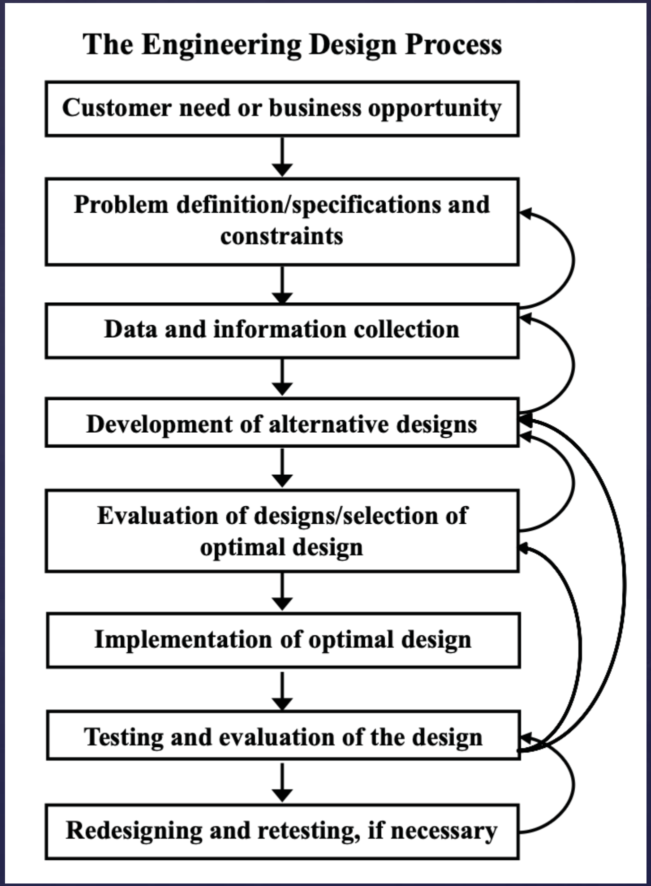
Steps 4 to 6: options, then a choice

Step 4 is plural, and it does not mean four finished products.

Development of alternative designs. Split the thing into its components, then lay out the real options for each. Some are either-or, e.g. battery or mains, steel or aluminum, one motor or two. Some are a range you sweep, e.g. wall thickness, gear ratio, clearance.

Evaluation and selection. CAD, stress analysis, computer modeling, materials, manufacturing. Your textbook also lists common sense and experience, which is unusually honest of it.

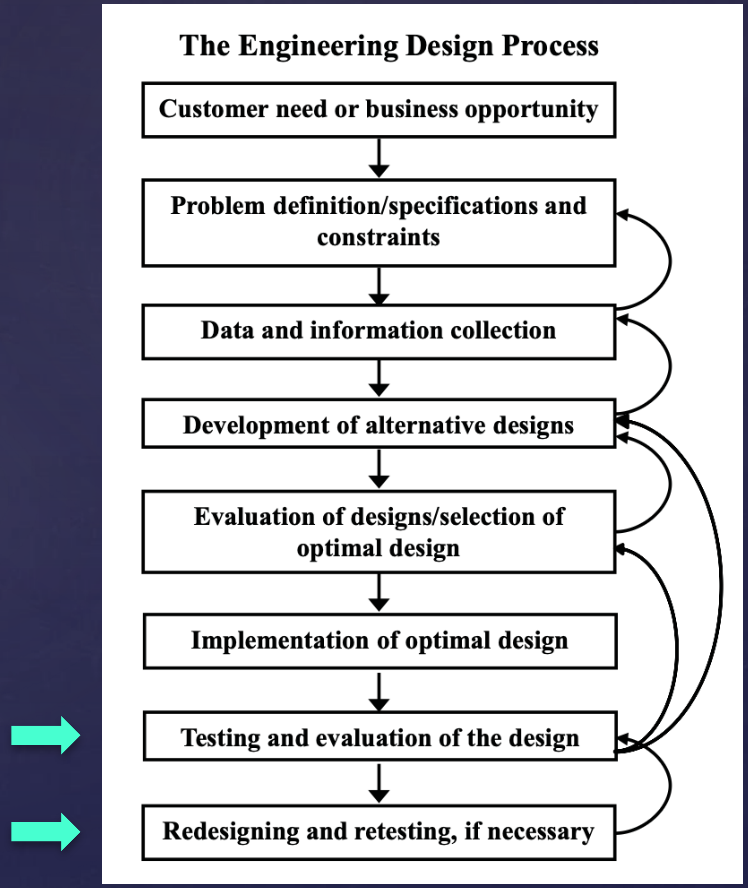
Implementation of the optimal design. Now you build the one you chose. Often more than one version, so you can test how well each actually meets the specifications. Or as you build the thing, you will see that you need to adapt or optimize based on unforeseen problems.



A mathematical model, a simulation, a breadboard, a development kit, a cardboard mock-up, all are used to get you to the finished thing.

Steps 7 and 8: test it, and expect to loop

Look again at the arrows curving backwards on the right of that diagram.



Testing and evaluation. Under realistic conditions first, then under worse ones. Find out how it fails before somebody else does.

Redesigning and retesting. Many iterations may be needed before a design meets the need, all of the specifications, and all of the constraints.

Specifications are not sacred. Once you know what is actually possible you may have to narrow them, or widen them.

A loop can be the whole product, like next year's phone, or one piece of it, like the antenna inside this year's.

Your phone is a list of specifications

Every feature on it was a number somebody had to choose.

Size and weight. Fixed early, and everything else had to fit inside them.

The display. Visible in daylight and in the dark, and adjustable between the two.

The battery. Life, safety and reliability, all of them in tension with size and weight.

Cost, and survival. Materials and manufacturing chosen on cost. It still has to survive being dropped, and still has to look worth buying.

And the parts nobody chooses. Radio emissions have to pass FCC limits here and EU limits there. The radio has to obey the Wi-Fi, Bluetooth and cellular standards or it talks to nothing. Even shipping the battery is regulated. None of that is negotiable.

Take something apart sometime and look. Your textbook recommends exactly that, and it is right.

Where needs come from

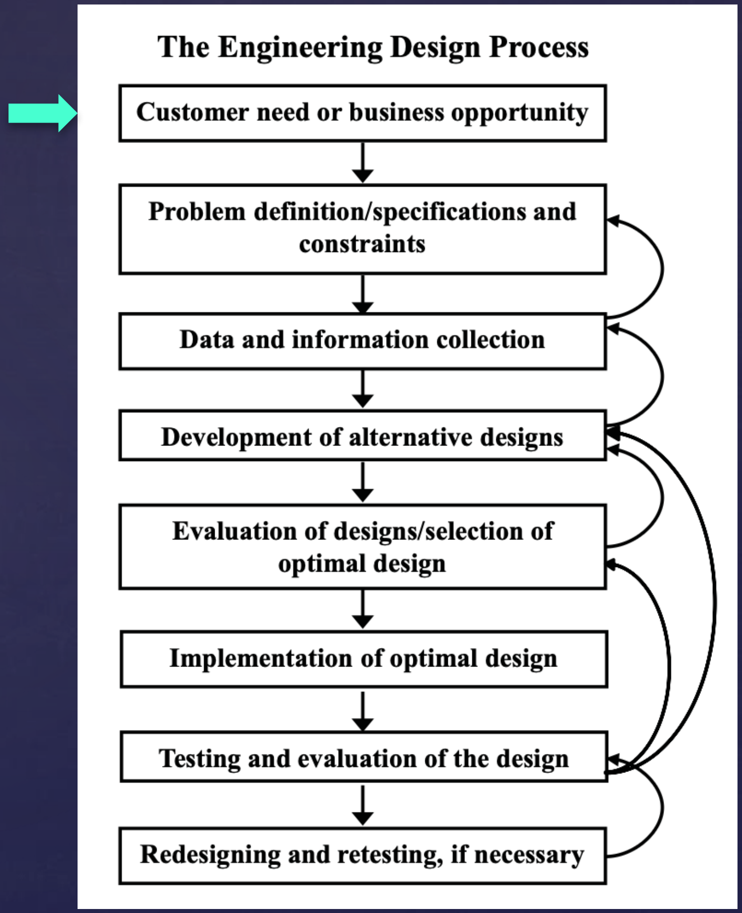
Step 1 is more important and less obvious than it sounds.

From a customer. The Air Force needs a missile that can put a 1,000-pound communications satellite into synchronous orbit. A need with a name attached.

From inside. Nobody asked 3M for a small yellow rectangle that sticks to things and then comes off again. No customer requested the Post-it.

From people who cannot tell you. Sometimes the people with the problem cannot describe it, or do not know they have it. This is why some design methods open with interviews and with watching people work.

From the world changing. A law moves, a standard is superseded, or a component you depend on stops being made. Nothing about your product changed, and it still needs redesigning.



Gamera: the problem

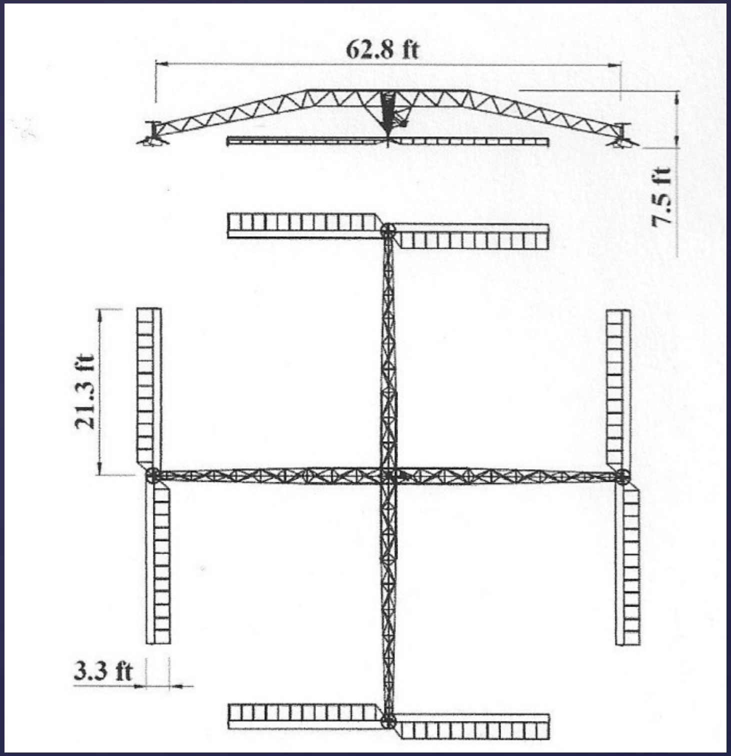
University of Maryland, 2008. A helicopter that flies on one person's muscle power.

The prize. Offered by the American Helicopter Society in 1980. $250,000. Unclaimed for twenty-eight years.

The rules were the specifications. Sixty seconds aloft. Three meters up, at least momentarily. Inside a ten-meter square. Human power only.

The state of the art. Seven seconds in 1989. Nineteen and a half in 1994. Nobody had been close.

Step 3 came first. No data existed for rotors that large turning that slowly, so the team ran its own test program before designing anything.



62.8 ft across, and 101 pounds

Gamera: what actually happened

The engineering story. Learning, striving, reaching milestones, repeat.

Gamera I. 11.4 seconds. A United States record, and nowhere near the prize. Retired.

Step 8, in public. They redesigned. Gamera II flew 49.9 seconds in June 2012, a world record.

September 2012. It reached 9.4 feet and crashed. The prize required 9.84 feet. Five inches short.

2013. A Canadian team, AeroVelo, won the prize after eighteen months of work.

They ran the process properly and still did not win it. Was the project worth doing?

This is not the only model

Other fields draw the same work differently. The vocabulary changes; the questions underneath do not.

Engineering design process

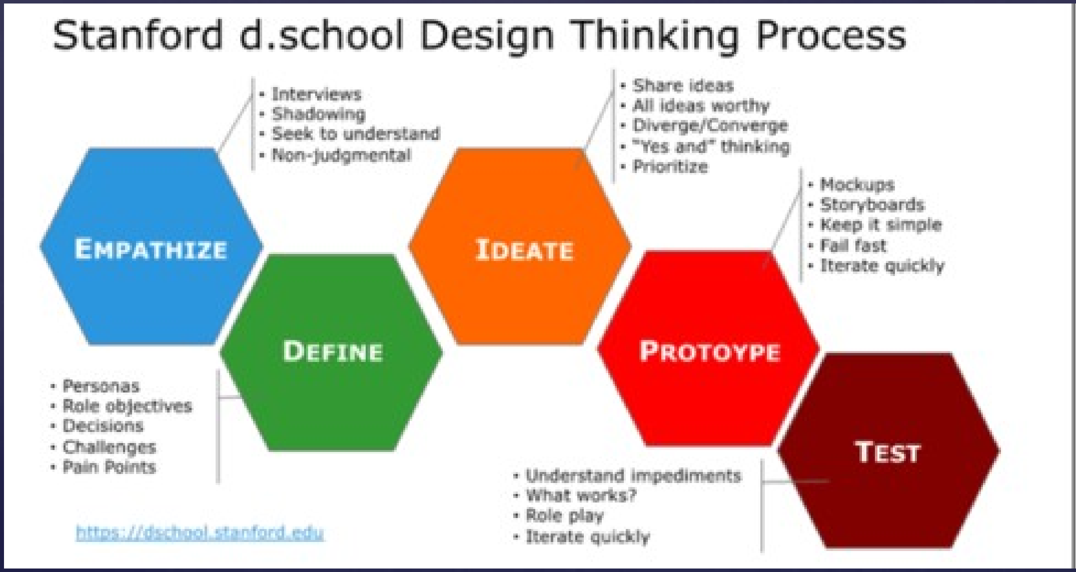
Eight steps. Strong on the numbers: specifications, constraints, and a defensible choice between options.

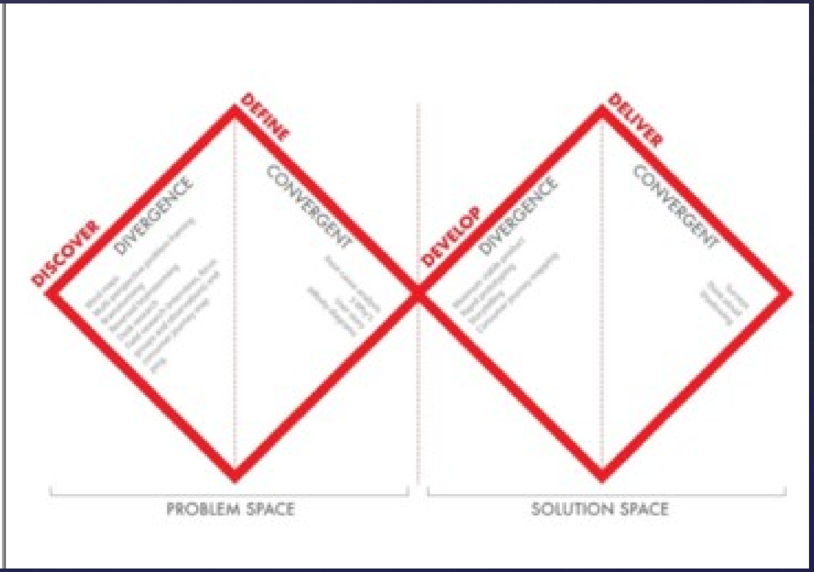
Assumes you can state the problem.

Design thinking (d.school)

Strong at the front: interviews, shadowing, finding out what people actually need before defining anything.

Assumes the problem is the hard part.





All three are asking the same two questions. Are we solving the right problem, and is this the right solution?

What your textbook leaves out

It is a book about succeeding as an engineering student. Design gets about ten pages out of three hundred and sixty.

Every industry has its own rules. A bridge answers to building codes and seismic design. An aircraft, a medical device and a chemical plant each answer to something different. Find out what yours are before you design for it.

Safety when it fails. Not whether it works, but what happens when it does not. In automotive that is ISO 26262, which grades each hazard and then dictates how you are permitted to design for it.

Security, and now AI. Both are becoming design requirements rather than afterthoughts. Nobody wants a backdoor in a brake controller, and the rules for AI are still being written.

The umbrella word. Engineers call these the "-ilities": safety, reliability, security, maintainability, manufacturability. Cheap to plan for, expensive to add late.

No textbook is complete. Notice what yours does not say, and ask where the rest lives.